

## DIPOLE TYPE BY LENGTH

**STEP 0 — always start here**  $\lambda = \text{wavelength (m), } l = \text{antenna rod length (m), } c = 3 \times 10^8 \text{ m/s}$

| $\lambda = \frac{c}{f} \text{ (m)}$ | ratio $\frac{l}{\lambda} \rightarrow$ | pick row below                                                |
|-------------------------------------|---------------------------------------|---------------------------------------------------------------|
| $l/\lambda$                         | Type                                  | Use formula                                                   |
| $< 1/50$                            | Hertzian (short)                      | HERTZIAN $S(R, \theta)$ , $D = 1.5$                           |
| $= 1/4$                             | Quarter-wave                          | ARB formula, $\pi l/\lambda = \pi/4$                          |
| $= 1/2$                             | Half-wave                             | half-wave shortcut, $D = 1.64$ , $R_{\text{rad}} = 73 \Omega$ |
| $= 1$                               | Full-wave                             | ARB, $D = 2.41$ , $R_{\text{rad}} = 199 \Omega$               |
| other                               | Arbitrary                             | ARB formula                                                   |

If  $l/\lambda < 1/50$  STOP and use Hertzian. Do NOT plug small  $l$  into the arbitrary formula — wrong shape.

## HERTZIAN DIPOLE — $S(R, \theta)$

|                                                                                                                                                               |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Far-field E,H phasors <small>small <math>l \ll \lambda</math></small>                                                                                         |
| $\vec{E}_\theta = \frac{jI_0 k l \eta_0}{4\pi R} e^{-jkR} \sin \theta$ , $\vec{H}_\phi = \vec{E}_\theta / \eta_0$                                             |
| Power density $l/\lambda < 1/50$ ; far field                                                                                                                  |
| $S(R, \theta) = \frac{\eta_0 k^2 I_0^2 l^2}{32\pi^2 R^2} \sin^2 \theta = S_{\text{max}} \sin^2 \theta \text{ (W/m}^2\text{)}$                                 |
| $l$ rod length (m); $I_0$ peak current (A); $R$ obs. dist. (m); $\theta$ from dipole axis; $k = 2\pi/\lambda$ (rad/m); $\eta_0 = 120\pi \approx 377 \Omega$ . |
| Max @ broadside $\theta_{\text{max}} = \pi/2$ ; $D$ exact $D = 1.5$ (1.76 dB)                                                                                 |
| Total radiated power $P_{\text{rad}} = \frac{\eta_0 \pi}{3} (I_0 l/\lambda)^2 \text{ (W)}$                                                                    |

## ARBITRARY-LENGTH DIPOLE — $S(\theta)$

|                                                                                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Power density at $R$ any $l$                                                                                                                                                                        |
| $S(\theta) = \frac{15 I_0^2}{\pi R^2} \left[ \frac{\cos(\frac{\pi l}{\lambda} \cos \theta) - \cos(\frac{\pi l}{\lambda})}{\sin \theta} \right]^2$                                                   |
| At $\theta = 0, \pi$ : bracket is 0/0. L'Hôpital $\Rightarrow S = 0$ (no axial radiation). TI-36X Pro alt: evaluate bracket at $\theta = 0.001 \text{ rad} \rightarrow \text{tiny} \rightarrow 0$ . |
| Normalized pattern <small>dimensionless</small> $F(\theta) = S(\theta)/S_{\text{max}}$                                                                                                              |
| Quarter-wave $l = \lambda/4$ $\pi l/\lambda = \pi/4$ , $\cos(\pi/4) = \sqrt{2}/2$                                                                                                                   |
| $\frac{S(\theta)}{S_0} = \left[ \frac{\cos(\frac{\pi}{4} \cos \theta) - \frac{\sqrt{2}}{2}}{\sin \theta} \right]^2$ , $S_0 = \frac{15 I_0^2}{\pi R^2}$                                              |
| $S_{\text{max}} = S(\pi/2) = S_0 \left(1 - \frac{\sqrt{2}}{2}\right)^2 \approx 0.0858 S_0$                                                                                                          |
| $D = 1.64$ , $R_{\text{rad}} = 36.5 \Omega$ , $\theta_{\text{max}} = \pi/2$ .                                                                                                                       |
| Half-wave $l = \lambda/2$ $\pi l/\lambda = \pi/2$ , <small>shortcut form</small>                                                                                                                    |
| $S(\theta) = \frac{15 I_0^2}{\pi R^2} \frac{\cos^2(\frac{\pi}{2} \cos \theta)}{\sin^2 \theta}$                                                                                                      |
| $D = 1.64$ , $R_{\text{rad}} = 73 \Omega$ , $\theta_{\text{max}} = \pi/2$ .                                                                                                                         |

## COMMON ANGLES (deg $\leftrightarrow$ rad, trig values)

|                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------|
| Conversion <small>multiply by <math>\pi/180</math> or <math>180/\pi</math></small>                                                  |
| $\theta_{\text{rad}} = \theta_{\text{deg}} \cdot \frac{\pi}{180}$ $\theta_{\text{deg}} = \theta_{\text{rad}} \cdot \frac{180}{\pi}$ |
| $^\circ$ rad $\sin \theta$ $\cos \theta$ $\sin^2 \theta$ $\cos^2 \theta$                                                            |
| 0 0 0 1 0 1                                                                                                                         |
| 30 $\pi/6$ 1/2 $\sqrt{3}/2$ 1/4 3/4                                                                                                 |
| 45 $\pi/4$ $\sqrt{2}/2$ $\sqrt{2}/2$ 1/2 1/2                                                                                        |
| 60 $\pi/3$ $\sqrt{3}/2$ 1/2 3/4 1/4                                                                                                 |
| 90 $\pi/2$ 1 0 1 0                                                                                                                  |
| 180 $\pi$ 0 -1 0 1                                                                                                                  |

Antenna formulas usually take  $\theta$  in rad. RAD mode on calc.

## LINEAR ANTENNA ARRAY

|                                                                                                                            |
|----------------------------------------------------------------------------------------------------------------------------|
| Setup $N$ identical elements along axis, spacing $d$                                                                       |
| elem pos $z_i = id$ , $i = 0, \dots, N-1$ ; $A_i = a_i e^{j\psi_i}$                                                        |
| Equal phase ( $\psi_i = 0$ ) $\Rightarrow$ broadside beam ( $\theta_{\text{max}} = \pi/2$ ).                               |
| Progressive phase $\psi_i = i\alpha \Rightarrow$ steered beam at $\cos \theta_{\text{max}} = -\alpha/(kd)$ .               |
| Pattern multiplication <small><math>S_e =</math> single element pattern</small>                                            |
| $S(R, \theta, \phi) = S_e(R, \theta, \phi) F_a(\theta)$                                                                    |
| Array factor (general) $k = 2\pi/\lambda$                                                                                  |
| $F_a(\theta) = \left  \sum_{i=0}^{N-1} A_i e^{jkid \cos \theta} \right ^2$                                                 |
| Uniform $A_i = 1$ , equal phase <small>closed form</small>                                                                 |
| $F_a(\theta) = \frac{\sin^2(N\gamma/2)}{\sin^2(\gamma/2)}$ , $\gamma = kd \cos \theta$                                     |
| Peak: $F_a^{\text{max}} = N^2$ at $\gamma = 0$ (broadside, $\theta = \pi/2$ ). Normal-ize: $F_a^{\text{norm}} = F_a/N^2$ . |
| HPBW (broadside, uniform) <small>use <math>Nd</math>, not <math>(N-1)d</math></small>                                      |
| $\beta \approx 0.88 \frac{1}{\sqrt{Nd}}$ (rad) $= 50.8^\circ \cdot \frac{1}{\sqrt{Nd}}$                                    |
| Grating lobes appear if $d > \lambda$ (broadside). Avoid.                                                                  |

## DIPOLE

|                                                      |
|------------------------------------------------------|
| $l$ — antenna length (m)                             |
| $\lambda = c/f$ (m); $c = 3 \times 10^8 \text{ m/s}$ |
| $I_0$ — peak current (A)                             |
| $k = 2\pi/\lambda$ (rad/m)                           |
| $R$ — obs. dist. (m); $\theta$ from axis             |
| $\eta_0 = 120\pi \approx 377 \Omega$                 |
| $S(R, \theta)$ — pwr density (W/m <sup>2</sup> )     |

## BEAM

|                                           |
|-------------------------------------------|
| $F(\theta) = S/S_{\text{max}}$ (unitless) |
| $a$ — coef. on $\theta^2$ in Gaussian     |
| $I_0$ — HPBW (current or $^\circ$ )       |
| $\Omega_p$ — pattern solid angle (sr)     |
| $D = 4\pi/\Omega_p$ — directivity         |
| $\xi$ — rad. efficiency; $G = \xi D$      |
| $D_{\text{dB}} = 10 \log_{10} D$          |

## FRIIS / dB

|                                                  |
|--------------------------------------------------|
| $P_t, P_r$ — TX/RX power (W)                     |
| $G_t, G_r$ — linear gains                        |
| $S_r = G_t P_t / (4\pi R^2)$ (W/m <sup>2</sup> ) |
| $P_r = G_t G_r (\lambda/4\pi R)^2 P_t$           |
| $G = 10^{G_{\text{dB}}/10}$                      |
| 3 dB = $\times 2$ , 10 dB = $\times 10$          |
| $P_N = k_B T_{\text{sys}} B$ (W)                 |

## APERTURE

|                                                |
|------------------------------------------------|
| $A_p$ — physical area (m <sup>2</sup> )        |
| $A_e = \lambda^2 D / (4\pi)$ (m <sup>2</sup> ) |
| $A_e \approx A_p$ for aperture                 |
| $D \approx 4\pi A_p / \lambda^2$ (unitless)    |
| $\beta \approx 0.88 \lambda / \ell$ (rect/sq.) |
| $\beta \approx 1.02 \lambda / d$ (circular)    |
| $2A_p \Rightarrow 2D$ , $\beta/\sqrt{2}$       |

## ARRAY

|                                                 |
|-------------------------------------------------|
| $N$ — number of elements                        |
| $d$ — spacing (m)                               |
| $\gamma = kd \cos \theta$                       |
| $F_a = \sin^2(N\gamma/2) / \sin^2(\gamma/2)$    |
| $F_a^{\text{max}} = N^2$ at $\theta = \pi/2$    |
| $F_a^{\text{norm}} = F_a/N^2$                   |
| $\beta \approx 0.88 \lambda / (Nd)$ (broadside) |

## BEAM PARAMETERS — $\beta$ , $\Omega_p$ , $D$

|                                                                                |
|--------------------------------------------------------------------------------|
| Normalize <small>always start here</small>                                     |
| $F(\theta) = S(\theta)/S_{\text{max}}$ (unitless)                              |
| Beamwidth $\beta$ at threshold $X$                                             |
| Set $F(\theta_X) = X$ , solve for $\theta_X$ ; $\beta = 2\theta_X$ (symmetric) |
| Threshold $X = 10^{\text{dB}/10}$ $\theta_X$ (Gauss. short-cut)                |
| HPBW (−3 dB) 0.5 $\sqrt{\frac{\ln 2}{a}}$                                      |
| −6 dB 0.25 $\sqrt{\frac{\ln 4}{a}}$                                            |
| −10 dB 0.1 $\sqrt{\frac{\ln 10}{a}}$                                           |
| any $F(\theta_X)$ any $X$ $F(\theta_X) = X$                                    |

|                                                                                                                                           |
|-------------------------------------------------------------------------------------------------------------------------------------------|
| Pattern solid angle $\Omega_p$ <small>recognize pattern, look up</small>                                                                  |
| $\Omega_p = \int_0^{2\pi} \int_0^\pi F(\theta) \sin \theta d\theta d\phi$ (sr)                                                            |
| No $\phi$ dep.: $\Omega_p = 2\pi \int_0^\pi F(\theta) \sin \theta d\theta$ .                                                              |
| $\theta$ is the integration variable — DON'T plug in a number for $\theta$ (e.g. $\theta_{1/2}$ ). Integrate over $\theta \in [0, \pi]$ . |

| Pattern $F(\theta)$           | $\Omega_p$ (sr)                     |
|-------------------------------|-------------------------------------|
| Isotropic ( $F=1$ )           | $4\pi \approx 12.57$                |
| Hertzian $\sin^2 \theta$      | $8\pi/3 \approx 8.38$               |
| Half-wave dipole              | $\approx 7.66$                      |
| Narrow Gauss $e^{-a\theta^2}$ | $\pi/a = \pi\theta_{1/2}^2 / \ln 2$ |
| 2-axis HPBW (aperture)        | $\beta_{xz} \cdot \beta_{yz}$       |
| Other (use integral)          | numerical                           |

TI-36X Pro: [2nd] [d/dx] for  $\int_{\square}^{\square}$ , RAD mode, multiply by  $2\pi$ .

|                                                                                                                                                                                            |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Directivity $D$ <small>always <math>D = 4\pi/\Omega_p</math>; common shortcuts below</small>                                                                                               |
| $D = \frac{4\pi}{\Omega_p}$ (unitless) $D_{\text{dB}} = 10 \log_{10} D$                                                                                                                    |
| WORD TRAP: “direction of max radiation” = angle $\theta_{\text{max}}$ (rad/deg). “directivity” = unitless number $D$ . Different things!                                                   |
| Dipoles $\rightarrow$ COMMON DIPOLE PARAMS table. Aperture: $D \approx 4\pi A_p / \lambda^2$ . 2-axis HPBW: $D \approx 4\pi/(\beta_{xz}\beta_{yz})$ . Circular: $D \approx 4\pi/\beta^2$ . |
| Gain (if asked) $\xi = \text{rad. eff.}$ , $0 < \xi \leq 1$                                                                                                                                |
| $G = \xi D$ $G_{\text{dB}} = 10 \log_{10} G$                                                                                                                                               |
| Lossless / ideal antenna $\Rightarrow G = D$ .                                                                                                                                             |

## COMMON DIPOLE PARAMETERS

| Type        | $l$               | $D$  | $D_{\text{dB}}$ | $R_{\text{rad}}$ ( $\Omega$ ) |
|-------------|-------------------|------|-----------------|-------------------------------|
| Isotropic   | N/A               | 1    | 0               | —                             |
| Hertzian    | $< \lambda/50$    | 1.5  | 1.76            | $80\pi^2 (l/\lambda)^2$       |
| Half-wave   | $\lambda/2$       | 1.64 | 2.15            | 73.2                          |
| $\lambda/4$ | $\lambda/4$ (gnd) | 1.64 | 2.15            | 36.6                          |
| monopole    |                   |      |                 |                               |
| Full-wave   | $\lambda$         | 2.41 | 3.82            | 199                           |

Half-wave:  $P_{\text{rad}} = 36.6 I_0^2 W$ .  $\lambda/4$  monopole (above ground): same patt. as half-wave but  $P_{\text{rad}}, R_{\text{rad}}$  halved. Lossless antenna ( $\xi = 1$ )  $\Rightarrow G = D$ . So in Friis, use  $G_t, G_r =$  these  $D$  values (e.g. half-wave  $G = 1.64$ ).

## EFFECTIVE AREA $A_e$

|                                                                                          |
|------------------------------------------------------------------------------------------|
| Definition <small>antenna's RX cross section (m<sup>2</sup>)</small>                     |
| $A_e = \frac{\lambda^2 D}{4\pi}$ (m <sup>2</sup> )                                       |
| Physical cross section <small>wire dipole, diameter <math>d</math></small>               |
| $A_p = l d = \frac{\lambda}{2} d$ (half-wave)                                            |
| Inverse: $D$ from $A_e$                                                                  |
| $D = \frac{4\pi A_e}{\lambda^2}$ (unitless)                                              |
| Hertzian special case $D = 1.5$                                                          |
| $A_e^{\text{Hertz}} = \frac{3\lambda^2}{8\pi} \approx 0.119 \lambda^2$ (m <sup>2</sup> ) |
| Thin wire: $A_e \gg A_p$ . Antenna captures field over $\sim \lambda$ .                  |

## FRIIS TRANSMISSION FORMULA

Pwr density at RX, then RX pwr  $G_t, G_r$  linear gains;  $\lambda = c/f$

$$S_r = \frac{G_t P_t}{4\pi R^2} \text{ (W/m}^2\text{); } P_r = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 P_t \text{ (W)}$$

Equivalent:  $P_r = S_r A_{e,r}$ .

Solve for  $R$  max range

$$R = \frac{\lambda}{4\pi} \sqrt{\frac{G_t G_r P_t}{P_r}} \text{ (m)}$$

Equivalent form using  $A_e$  recv. area form

$$P_r = \frac{G_t P_t A_{e,r}}{4\pi R^2}, \quad G_r = \frac{4\pi A_{e,r}}{\lambda^2}$$

Free-space path loss:  $L_{fs} = (4\pi R/\lambda)^2$ , so  $P_r = G_t G_r P_t / L_{fs}$ .

General form (off-axis) when patterns NOT peak-aligned

$$P_r = G_t G_r \left( \frac{\lambda}{4\pi R} \right)^2 P_t F_t(\theta_t, \phi_t) F_r(\theta_r, \phi_r)$$

$F_t, F_r$  normalized patterns at the link directions;  $F = 1$  when aligned.

Recipe 1.  $\lambda = c/f$ . 2. Convert dB gains to linear:  $G = 10^{G_{\text{dB}}/10}$ .

3. If TX is “half-wave dipole” use  $G_t = 1.64$ . 4. Plug into Friis.

Use linear  $G$ , not dB. Convert FIRST.

## dB $\leftrightarrow$ LINEAR

Works for any power-like quantity:  $G$  gain,  $D$  directivity,  $P_r/P_t$ , SNR, etc.

$$X_{\text{dB}} = 10 \log_{10} X_{\text{lin}} \quad X_{\text{lin}} = 10^{X_{\text{dB}}/10}$$

e.g.  $D_{\text{dB}} = 10 \log_{10} D$ ;  $G_{\text{dB}} = 10 \log_{10} G$ ;  
 $SNR_{\text{dB}} = 10 \log_{10} (P_r/P_N)$ .

| dB | linear | dB  | linear |
|----|--------|-----|--------|
| 0  | 1      | 10  | 10     |
| 3  | 2      | 13  | 20     |
| 6  | 4      | 20  | 100    |
| 7  | 5      | 30  | 1000   |
| −3 | 0.5    | −10 | 0.1    |

Memorize: 3 dB  $\Leftrightarrow \times 2$ , 10 dB  $\Leftrightarrow \times 10$ . Add dB = multiply linear.

For E-field / amplitude use 20 log not 10 log (power  $\propto E^2$ ).

## NOISE & SNR

|                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------|
| Noise power <small>thermal noise into matched receiver</small>                                                  |
| $P_N = k_B T_{\text{sys}} B$ (W)                                                                                |
| $k_B = 1.38 \times 10^{-23} \text{ J/K}$ ; $T_{\text{sys}}$ system noise temp (K); $B$ receiver bandwidth (Hz). |
| Signal-to-noise ratio                                                                                           |
| $SNR = \frac{P_{\text{rec}}}{P_N}$ (unitless) $SNR_{\text{dB}} = 10 \log_{10} \frac{P_{\text{rec}}}{P_N}$       |
| Typical comm. link needs $SNR > 10 \text{ dB}$ for usable signal.                                               |

## APERTURE ANTENNAS (horn, dish)

Directivity (any shape, uniform illum.)

$$D \approx \frac{4\pi A_p}{\lambda^2} \text{ (unitless)} \Rightarrow A_e \approx A_p$$

Directivity from beamwidths *any aperture*

$$D \approx \frac{4\pi}{\beta_{xz}\beta_{yz}} \text{ (use rad); circular: } D \approx 4\pi/\beta^2$$

HPBW and  $D$  by aperture shape *uniform illum.;  $\beta$  in rad*

| Shape                        | $A_p$           | HPBW (rad)                                                                   | $D$ (unitless)                                                           |
|------------------------------|-----------------|------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Rect. $\ell_x \times \ell_y$ | $\ell_x \ell_y$ | $\beta_x \approx 0.88\lambda/\ell_x$<br>$\beta_y \approx 0.88\lambda/\ell_y$ | $D \approx 4\pi/(\beta_x \beta_y)$<br>$= 4\pi \ell_x \ell_y / \lambda^2$ |
| Square $\ell$                | $\ell^2$        | $\beta \approx 0.88\lambda/\ell$                                             | $D \approx \frac{4\pi}{\beta^2} = \frac{4\pi \ell^2}{\lambda^2}$         |
| Circular (diam. $d$ )        | $\pi d^2/4$     | $\beta \approx 1.02\lambda/d$                                                | $D \approx \frac{4\pi}{\beta^2} = \frac{\pi^2 d^2}{\lambda^2}$           |

Scaling rules  $D \propto A_p/\lambda^2$ ;  $\beta \propto \lambda/\sqrt{A_p}$

| Change                                          | New $D$                                 | New $\beta$                                        |
|-------------------------------------------------|-----------------------------------------|----------------------------------------------------|
| Double ( $A_p \rightarrow 2A_p$ )               | area $D_{\text{new}} = 2D_{\text{old}}$ | $\beta_{\text{new}} = \beta_{\text{old}}/\sqrt{2}$ |
| Double freq ( $\lambda \rightarrow \lambda/2$ ) | $D_{\text{new}} = 4D_{\text{old}}$      | $\beta_{\text{new}} = \beta_{\text{old}}/2$        |